What is Data Mining?
The Big Data Landscape and the KDD Process

BDAT 602 – Data Mining and Applications — Lecture 1

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Outline

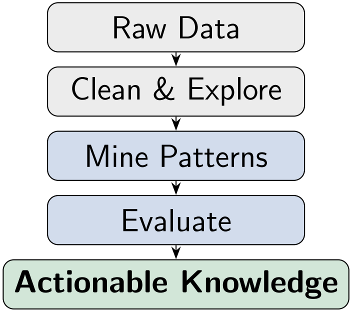
What This Course Is About

The central question:

> *"We have more data than ever before. How do we turn it into knowledge?"*

BDAT 602 answers this by teaching:

- How to clean and prepare large, messy datasets in R
- How to find hidden patterns — clusters, rules, and categories
- How to evaluate whether a discovered pattern is meaningful



Where You Are Coming From

Prior Knowledge We Build On

Students entering BDAT 602 already have experience with:

- Working in **R** and **RStudio** for data handling and visualisation
- Connecting R to **Apache Spark** via `sparklyr` for distributed computation
- The core definition and **4 Vs** of big data from prior coursework
- Basic data manipulation using `dplyr`, `readr`, and `ggplot2`

Where You Are Coming From

What is genuinely new in BDAT 602:

- Structured **discovery workflows**: KDD pipeline and CRISP-DM
- The three core mining families: **association rules, clustering, classification**
- Applying algorithms to real and large-scale data.
- Rigorous **evaluation and interpretation** of everything we find.

The Course at a Glance

Topic	**Core Algorithm(s)**	**Key R Tools**
What is Data Mining? KDD / CRISP-DM	Conceptual Foundations	`readr`, `data.table`
Data Cleaning and EDA at Scale	Preprocessing Techniques	`tidyverse`, `sparklyr`
Association Rule Mining	Apriori, FP-Growth	`arules`, `arulesViz`
Cluster Analysis	k-means, DBSCAN	`cluster`, `factoextra`
Classification and Evaluation	Trees, Naïve Bayes, kNN	`rpart`, `caret`, `pROC`
Text Mining, Streams & Capstone Lab	TF-IDF, LDA, Hoeffding	`tidytext`, `tm`

The Course at a Glance

One Dataset, Six Weeks

All lectures use a single **simulated health insurance portfolio** (500,000 records, 38 variables). Every new technique builds on familiar data — no time wasted re-learning context.

Defining Data Mining

Working Definition

Data mining is the computational process of discovering *non-trivial*, *previously unknown*, and *potentially useful* patterns and relationships in large datasets.

Key words unpacked:

- **Non-trivial** — not obvious by simple inspection
- **Previously unknown** — not hand-coded business rules
- **Potentially useful** — actionable in a real decision
- **Large datasets** — scale matters; patterns invisible at small n emerge at big n

Data mining is not:

- Writing an SQL `SELECT` query

  (that is data retrieval)
- Computing a mean or median

  (that is descriptive statistics)
- Testing a pre-formed hypothesis

  (that is confirmatory analysis)
- Simply plotting data

  (though EDA supports mining)

Data mining is:

- *Automated* search for structure
- *Exploratory* in nature
- Applied to data *at scale*

Data Mining vs. Related Fields

Field	Primary Question	Relationship to Data Mining
Statistics	Is this pattern real or due to chance?	Data Mining borrows statistical methods but places less emphasis on hypothesis testing and *p*-values.
Machine Learning	How can predictive models be built from data?	Data Mining uses machine learning algorithms for classification, prediction, and clustering.
Artificial Intelligence	How can human intelligence be simulated computationally?	Data Mining is one practical application area within Artificial Intelligence.
Database Systems	How can data be efficiently stored and retrieved?	Data Mining relies on databases as sources for large-scale structured data.
Data Science	How can raw data be transformed into actionable insight?	Data Mining represents the pattern-discovery and knowledge-extraction stage of Data Science.

The Question DM tries to Answer

What meaningful patterns, clusters, trends, and relationships exist within large datasets?

Key Clarification

Data mining is not a replacement for statistics or machine learning. It is a **process** that draws on methods from all these fields to discover knowledge at scale.

The Five Core Mining Tasks

1. **Association Rule Mining**
   "What things tend to occur together?"
   Customers who buy dental cover also tend to buy vision cover.
2. **Clustering**
   "What are the natural groups in the data?"
   Segmenting policyholders into low-, medium-, and high-risk groups.
4. **Regression / Prediction**
   "What numeric value should we expect?"
   How much will this policyholder claim next year?
5. **Classification**
   "What category does a new observation belong to?"
   Is this new claim likely to be fraudulent?

The Five Core Mining Tasks

1. **Anomaly / Outlier Detection**
   "Which observations are unusual or suspicious?"

   Detecting claims that deviate unexpectedly from the norm.

Supervised vs. Unsupervised

Association rules and clustering are **unsupervised** (no labels needed). Classification and regression are **supervised** (labelled outcomes required).

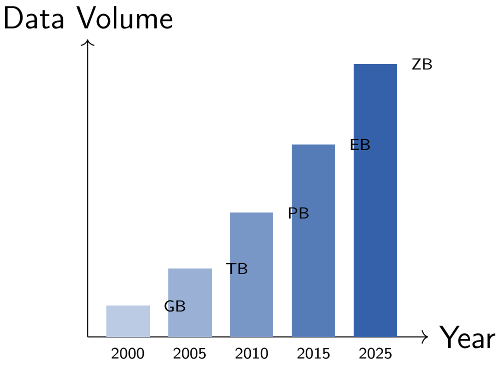
Real-World Applications of Data Mining

Industry	Application	Mining Task
Retail / E-commerce	"Customers who bought X also bought Y"	Association rules
Banking / Insurance	Detecting fraudulent transactions	Anomaly detection
Healthcare	Grouping patients by disease risk profile	Clustering
Telecommunications	Predicting which customers will cancel	Classification
Marketing	Customer segmentation for targeted offers	Clustering
Manufacturing	Predictive maintenance of machinery	Regression
Social Media	Detecting spam and fake accounts	Classification
Credit Scoring	Estimating probability of loan default	Classification
Epidemiology	Identifying disease spread patterns	Association / Cluste
This course	Health insurance portfolio analysis	**All of the above**

Why Big Data Changes Everything

The scale of data today:

- Over **2.5 quintillion bytes** of data created globally every day
- A single insurer may process **millions of claims** per year
- A hospital generates terabytes of records, imaging, and sensor data
- Social media platforms handle **billions of interactions** per hour



Why Big Data Changes Everything

Why traditional methods fail at this scale:

- A single machine's RAM cannot hold the dataset in memory
- Algorithms designed for $n < 100{,}000$ become unacceptably slow
- A `for` loop over 500 million rows in base R could run for *hours*
- Solution: **distributed computing** via Apache Spark + `sparklyr`

The 4 Vs of Big Data: A Deeper Look

1. **Volume** — *How much data?*
   Our dataset: 500,000 records. Industry datasets reach petabytes. Traditional tools break; distributed tools scale.
   Tool response: `sparklyr`, `data.table`

2. **Velocity** — *How fast is data arriving?*
   Insurance claims, card transactions, and sensor readings arrive in real-time. Fraud must be caught in milliseconds.
   Mining response: stream mining

3. **Variety** — *What types of data?*
   Our dataset contains numerics, categories, dates, free text (`complaint_notes`), and binary flags. Each type requires different treatment.
   Mining response: text mining, encoding

The 4 Vs of Big Data: A Deeper Look

1. **Veracity** — *How trustworthy is the data?*
   Our dataset intentionally includes missing values, outliers, and entry errors. Real data is never clean.

   Mining response: data cleaning pipeline

The 5th V — Value

All four Vs are irrelevant unless the mining process produces **actionable insight** that justifies the cost of infrastructure. Value is the ultimate goal.

Data Types in Mining: A Taxonomy

Structured data (most mining algorithms assume this):

- **Continuous / interval**: `age`, `income`, `bmi`, `claim_amount`
- **Ordinal**: `customer_rating` (1–5), `education` level
- **Binary**: `smoker`, `fraud_flag`, `churned`

Semi-structured and unstructured:

- **Free text**: `complaint_notes` column in our dataset — requires NLP tools.
- **Transactional**: records of individual purchases or events — required for market basket analysis.

Data Types in Mining: A Taxonomy

Structured data (most mining algorithms assume this):

- **Nominal / categorical**: `sex`, `region`, `plan_tier`, `payment_method`
- **Count**: `num_claims`, `num_visits`, `num_prescriptions`
- **Date / time**: `policy_start_date`, `last_claim_date`

Semi-structured and unstructured:

- **Time series**: sequences of measurements over time — stream mining
- **Spatial**: geographic coordinates — spatial hotspot detection
- **Network / graph**: relationships between entities — PageRank

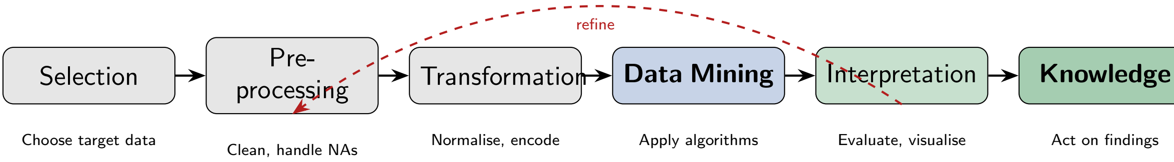
Why Data Types Matter

The data type of each variable determines which algorithms can be applied, how the variable must be encoded, and which distance or similarity measures are appropriate.

From Data to Knowledge: The KDD Pipeline

Knowledge Discovery in Databases (KDD)

KDD is the **complete process** of transforming raw data into interpretable, actionable knowledge. Data mining is one *step* within this larger process — not the whole thing.



The KDD process is **iterative**, not linear. Interpretation often sends us back to clean more data or transform variables differently.

Step	KDD Stage	What We Do	Tool in R
1	**Selection**	Load 500,000 insurance records and select relevant variables for analysis	`readr`, `data.table`
2	**Preprocessing**	Handle missing BMI and income values, remove or flag outliers, and deduplicate records	`tidyverse`, `skimr`
3	**Transformation**	Normalise premiums, encode `plan_tier`, and log-transform `income`	`recipes`, `DataExplorer`
4	**Data Mining**	Apply Apriori, *k*-means, and decision trees	`arules`, `cluster`, `rpart`
5	**Interpretation**	Visualise association rules, interpret clusters, and evaluate models using ROC/AUC	`arulesViz`, `pROC`, `ggplot2`
6	**Knowledge**	Present findings such as high-risk policyholders and frequently co-occurring products	`R Markdown`

KDD Step by Step: Applied to Our Dataset

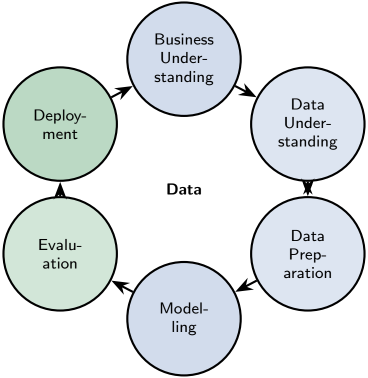
Each Week Maps to a KDD Stage

Weeks 1–2 cover Steps 1–3. Weeks 3–6 cover Steps 4–6. The capstone in Week 6 requires you to execute all six steps end-to-end.

CRISP-DM: The Industry Standard Process

Cross-Industry Standard Process for Data Mining (CRISP-DM) was developed in the late 1990s by a consortium of companies. It remains the **most widely used** data mining process model in industry today.

Key feature: arrows go in *both directions* — the process is cyclical, not a one-way waterfall.



CRISP-DM Phase by Phase: Applied to Our Insurance Project

Phase	What It Means	Our Course Example
1. Business Understanding	Define the problem and project objectives in business terms	Identify high-risk policyholders, detect fraud, and reduce customer churn
2. Data Understanding	Explore the dataset and assess data quality	Perform EDA using `skimr`, examine missing values, and analyse distributions
3. Data Preparation	Clean, transform, and prepare features for modelling	Impute missing BMI values, log-transform income, and encode `plan_tier`
4. Modelling	Apply and tune data mining or machine learning algorithms	Generate Apriori rules, perform k-means clustering, and build decision tree classifiers

CRISP-DM Phase by Phase: Applied to Our Insurance Project

Phase	What It Means	Our Course Example
5. Evaluation	Measure model quality against business objectives	Evaluate using lift, silhouette score, confusion matrix, and ROC/AUC metrics
6. Deployment	Present and operationalise insights for stakeholders	Create R Markdown reports, develop Shiny dashboards, and provide policy recommendations

CRISP-DM vs. KDD

KDD and CRISP-DM describe the same reality from different angles. KDD is more **technical** (algorithm-centric); CRISP-DM is more **business-facing** (goal-centric). In practice, you use both simultaneously.

Why Phase 1 (Business Understanding) Is Often the Hardest

A common mistake: jumping straight to algorithms without a clear question.

Example: the wrong approach

> *"Let us apply k-means to the insurance data and see what comes out."*

The right approach requires answering:

- What business *decision* will this mining result inform?
- Who is the end-user of these findings?

Three well-formed mining questions for our dataset:

1. *"Which pairs of add-on covers tend to be purchased together, and can we use this to design bundle offers?"* $\rightarrow$ Association rules
2. *"Can we group our 500,000 policyholders into risk segments to guide premium pricing?"* $\rightarrow$ Clustering

Why Phase 1 (Business Understanding) Is Often the Hardest

The right approach requires answering:

- What does "success" look like numerically?
- What are the consequences of a false positive vs. a false negative?
- Is the data I need actually available and of sufficient quality?

Three well-formed mining questions for our dataset:

1. *"Can we identify fraudulent claims before they are paid, using the claimant's profile?"* $\rightarrow$ Classification

The Course Dataset: Global Health Insurance Portfolio

Scenario

A simulated operational database of a multinational health insurer: **500,000 policy records** spanning 2023–2024, covering policyholders across 6 continental regions.

Variable groups (38 columns total):

- **Identifiers (4)**: record, policyholder, agent, provider IDs
- **Demographics (8)**: age, sex, BMI, smoker, region, education, employment, income
- **Fraud (1)**: `fraud_flag` — rare event (≈3%)
- **Behaviour (5)**: payment method, auto-pay, app logins, support calls, customer rating

The Course Dataset: Global Health Insurance Portfolio

Variable groups (38 columns total):

- **Policy details (10)**: plan tier, premium, deductible, coverage, policy age, 4 add-on riders
- **Medical usage (4)**: chronic conditions, visits, prescriptions, hospital admissions
- **Claims (5)**: number of claims, past claim, claim amount, days since last claim, weekend flag
- **Text (1)**: `complaint_notes` — free text
- **Targets (3)**: `churned`, `high_risk`, `claim_amount`

Built-in data quality issues for Week 2:

- 5% missing BMI; 4% missing income
- 10% missing complaint notes
- Injected extreme outliers in BMI and income

Generating the Dataset in R

```r
# Source the data generation function (provided on the course page)
source("simulate_bdat602.R")

# Generate 500,000 records at Big Data scale
health_data <- simulate_bdat602(n = 500000, seed = 602)

# Quick orientation
dim(health_data)           # 500000 x 40
names(health_data)         # inspect variable names
str(health_data)           # data types
head(health_data, 3)       # first 3 rows

# For experimentation during lectures, use a smaller slice
health_small <- simulate_bdat602(n = 10000, seed = 602)
```

Reproducibility Rule

The `seed = 602` argument ensures every student generates **identical data**. Always set a seed before any random process in research, teaching, or production code. A result that cannot be reproduced is not a result.

A First Look at the Data

```r
library(dplyr)
library(skimr)

# Summary statistics for every variable
skim(health_small)

# Count of each plan tier
health_small |>
count(plan_tier) |>
mutate(pct = round(100 * n / sum(n), 1))

# Distribution of claim amounts (our main continuous variable)
health_small |>
filter(claim_amount > 0) |>
summarise(
mean_claim   = mean(claim_amount),
median_claim = median(claim_amount),
```

```r
sd_claim       = sd(claim_amount),
max_claim      = max(claim_amount),
pct_zero       = mean(health_small$claim_amount == 0) * 100
)

# Fraud prevalence (rare event)
health_small |>
summarise(fraud_rate = mean(fraud_flag) * 100)
```

Loading Big Data into Spark

```
library(sparklyr)
library(dplyr)

# Connect R to a local Spark session
sc <- spark_connect(master = "local[*]")

# Copy the full 500,000-row dataset to Spark
health_tbl <- copy_to(
sc,
health_data,
name      = "health_insurance",
overwrite = TRUE
)

# Spark does not load data into R memory; computation stays
    distributed
health_tbl |> count()              # count rows in Spark
```

```r
health_tbl |> glimpse()          # inspect schema

# Take a 10% Bernoulli sample for exploratory work
health_sample <- health_tbl |>
sdf_sample(fraction = 0.10, replacement = FALSE, seed = 602)

sdf_nrow(health_sample)          # should be ~50,000
```

Remember

When you use `sparklyr`, data lives in Spark — not in R's memory. Commands like `filter()`, `mutate()`, and `group_by()` are translated to Spark operations. Only `collect()` brings results back to R.

The R Ecosystem for Data Mining

Data ingestion and manipulation:

- `readr` — fast reading of CSV and delimited files
- `data.table` — memory-efficient processing of large flat files
- `dplyr` — readable data wrangling grammar
- `sparklyr` — distributed processing via Apache Spark

Mining algorithms:

- `arules` + `arulesViz` — association rule mining
- `cluster` + `factoextra` — clustering and visualisation
- `dbscan` — density-based clustering
- `rpart` + `e1071` — decision trees and Naïve Bayes
- `caret` + `pROC` — model training and evaluation

The R Ecosystem for Data Mining

Exploration and cleaning:

- `skimr` — rapid summary statistics for all variable types
- `DataExplorer` — automated EDA report generation
- `janitor` — cleaning column names, duplicates, tabulations
- `tidyr` — reshaping and pivoting data frames

Mining algorithms:

- `tm` + `tidytext` — text mining

Reporting:

- `R Markdown` — reproducible reports and capstone submission
- `ggplot2` — publication-quality visualisation

R Orientation: Reading and Inspecting Data

```r
library(readr)
library(data.table)
library(dplyr)

# --- Option A: readr (tidyverse style, moderate files) ---
df_readr <- read_csv("health_insurance.csv")

# --- Option B: data.table (fastest for large flat files) ---
df_dt <- fread("health_insurance.csv")

# --- Inspect data types ---
glimpse(df_readr)

# --- Check for missing values ---
colSums(is.na(df_readr))
```

```r
# --- Frequency table for a nominal variable ---
df_readr |>
count(region, sort = TRUE) |>
mutate(pct = round(100 * n / sum(n), 1))

# --- Quick numeric summary ---
df_readr |>
select(age, bmi, income, claim_amount) |>
summary()
```

R Orientation: First Exploratory Visualisations

```r
library(ggplot2)

# Distribution of claim amounts (highly right-skewed)
ggplot(health_small |> filter(claim_amount > 0),
aes(x = claim_amount)) +
geom_histogram(bins = 60, fill = "steelblue", colour = "white") +
scale_x_log10(labels = scales::comma) +
labs(title = "Distribution of Claim Amounts (log scale)",
x = "Claim Amount (USD, log scale)", y = "Count") +
theme_minimal()
```

R Orientation: First Exploratory Visualisations

```r
# Claim amount by plan tier (box plot)
ggplot(health_small |> filter(claim_amount > 0),
aes(x = plan_tier, y = claim_amount, fill = plan_tier)) +
geom_boxplot(outlier.alpha = 0.2) +
scale_y_log10(labels = scales::comma) +
labs(title = "Claim Amount by Plan Tier",
x = "Plan Tier", y = "Claim Amount (log scale)") +
theme_minimal() +
theme(legend.position = "none")
```

R Orientation: Automated EDA with DataExplorer

```r
library(DataExplorer)

# One-line automated EDA report (HTML output)
create_report(
health_small,
output_file   = "eda_report.html",
output_dir    = ".",
report_title  = "BDAT 602: Health Insurance EDA"
)

# --- Or explore interactively ---

# Missing value map
plot_missing(health_small)
```

R Orientation: Automated EDA with DataExplorer

```
# Distribution of all continuous variables
plot_histogram(health_small)
# Correlation heatmap (numeric variables only)
plot_correlation(
health_small |> select(age, bmi, income, premium,
claim_amount, num_claims,
num_chronic_conditions)
)
```

What We Covered Today

Conceptual foundations:

- ✓ A precise definition of data mining and what distinguishes it from statistics, ML, and database queries
- ✓ The five core mining tasks: association rules, clustering, classification, regression, anomaly detection
- ✓ The 4 Vs of big data and why they demand new tools
- ✓ Data types in mining: continuous, categorical, ordinal, text, transactional

Practical (R):

- ✓ Generating the course dataset with `simulate_bdat602()`
- ✓ Reading large files with `readr` and `data.table`
- ✓ Loading data into Spark with `sparklyr`
- ✓ First inspection with `glimpse()`, `skim()`, and `summary()`

What We Covered Today

Conceptual foundations:

- ✓ The KDD pipeline: six steps from raw data to knowledge
- ✓ CRISP-DM: the industry-standard process model
- ✓ Why Business Understanding (Phase 1) is the hardest and most important step

Practical (R):

- ✓ Automated EDA with `DataExplorer`
- ✓ First visualisations: histograms and box plots with `ggplot2`

Key Takeaway

Data mining is a **process**, not a single algorithm. The quality of the process determines the quality of the knowledge produced.

Looking Ahead: Lecture 2

Next Week: Getting Data Ready — Cleaning, Transforming, and Exploring at Scale

Before any algorithm can run, the data must be trustworthy. Week 2 is entirely dedicated to the craft of data preparation — often the most time-consuming step in any real project.

Key topics in Lecture 2:

- Diagnosing and handling missing values (imputation strategies)
- Detecting and treating outliers in BMI and income

Dataset continuity:

- Same `health_data` object — we begin treating the data quality issues injected in Week 1

Looking Ahead: Lecture 2

Key topics in Lecture 2:

- Normalisation and standardisation: why and when
- Encoding categorical variables for mining algorithms
- PCA as a dimensionality reduction tool (intuition only)
- Scaling EDA pipelines with `sparklyr`

Dataset continuity:

- Focus on preparing `bmi`, `income`, `plan_tier`, and `complaint_notes` for downstream mining

R tools:

- `tidyverse`, `DataExplorer`, `skimr`
- `sparklyr` for distributed cleaning
- `recipes` (preview for Week 5)

Key Takeaways

Three Rules for Every Data Mining Project

1. **Understand the business question first.** An algorithm applied without a clear goal produces output, not insight. Ask: what decision will this result support?
2. **Data quality determines result quality.** No algorithm can compensate for dirty, incomplete, or biased data. Invest heavily in Steps 1–3 of KDD before Step 4.
3. **Interpretation is not optional.** A cluster, a rule, or a classified label is meaningless until it is explained in terms a domain expert can act on.

Mining Task	Typical Question	Week
Association Rules	Which products co-occur?	3
Clustering	What are the natural segments?	4
Classification	Which category does this belong?	5
Text Mining	What topics appear in complaints?	6

Self-Study and Further Reading

Core reading for Lecture 1:

- Han, Kamber & Pei (2011), *Data Mining: Concepts and Techniques* (3rd ed.):
  Ch. 1: Introduction
  Ch. 2: Data (types, quality, preprocessing overview)
- Tan, Steinbach & Kumar (2019), *Introduction to Data Mining* (2nd ed.):
  Ch. 1: Introduction
- Leskovec, Rajaraman & Ullman (2020), *Mining of Massive Datasets*:
  Ch. 1: Data Mining (free at `mmds.org`)

R resources:

- Wickham & Grolemund (2017), *R for Data Science*:
  Ch. 1–5: R basics, `dplyr`, visualisation
  Free at `r4ds.had.co.nz`
- `sparklyr` documentation:
  `spark.rstudio.com`

Practice exercises:

- Generate the dataset and run `create_report()` from `DataExplorer`. Write three observations about the data.
- Identify which of the 5 mining tasks is most relevant to each of the three business questions on the CRISP-DM slide.
- Load the dataset into Spark and count rows by `region` using `sparklyr`.

Questions?

Next: Lecture 2 — Getting Data Ready:
Cleaning, Transforming, and Exploring at Scale

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